STUDENT SKILLS AND COURSE RECOMMENDATION SYSTEM

Research Paper 2: Using Artificial Intelligence to Match Student Interests with Learning Opportunities

Student Research Paper | October 2026

Abstract

Artificial intelligence can help organize information about student interests, learning goals, and course content. This paper examines a possible AI-supported recommendation system that uses course descriptions and student profiles to identify relevant learning opportunities. It discusses keyword and skill extraction, course matching, recommendation explanations, and the need for human review. The design is proposed for educational use and does not claim to have been implemented or experimentally validated. Its central argument is that AI recommendations should be understandable, current, and supportive of student choice.

Keywords: artificial intelligence, course matching, student interests, NLP, explainable recommendations

1. Introduction

Educational websites and institutions provide many courses, but the amount of information can make selection difficult. A student may know that they enjoy science, design, business, or computing without knowing which course best matches those interests. Artificial intelligence (AI) can assist by finding relationships between a student profile and the information contained in course descriptions. Used carefully, this can make course discovery more efficient.

2. Research Problem

Traditional course searches often depend on broad categories or exact keywords. They may miss a suitable course when it uses different terminology from the student's search. A system that understands related skills and learning outcomes could provide more useful matches. At the same time, automated systems may misunderstand vague descriptions or recommend courses based on incomplete information.

3. Aim and Objectives

This paper aims to outline an AI-supported course recommendation model. Its objectives are to examine how student interests can be represented, explain how course descriptions can be processed, propose a matching method, and identify safeguards that make recommendations useful and fair.

4. System Design

The proposed system contains four main components: a student profile module, a course information database, a matching engine, and an explanation interface. The profile may include stated interests, completed learning, desired difficulty, and goals. The database should record course topics, prerequisites, outcomes, duration, and skills developed. The matching engine compares the two sets of information, while the interface shows the reasons behind each result.

5. Role of Natural Language Processing

Natural Language Processing (NLP) can help extract important terms from course descriptions, such as programming, communication, data analysis, teamwork, or problem-solving. Related terms can be grouped under

a shared skill label. For example, a course description that says learners will interpret datasets may be connected to data-analysis skills. Such extraction should be reviewed because the same word can have different meanings in different subjects.

6. Recommendation and Explanation

A practical model can combine several signals: match with stated interests, skills taught, prerequisite suitability, learning goals, and preferred level. The system can rank courses using a weighted score, but it should also show a plain-language explanation such as: “Recommended because this course teaches introductory data analysis and matches your stated interest in science.” Students should be able to change preferences, report inaccurate information, and explore courses outside the first list.

7. Related Research

Prior work provides examples of different recommendation approaches. Vo and colleagues describe an NLP-based system for identifying technical skills and supporting learning-path decisions in computer science and IT. Beutling and Spahic-Bogdanovic connect learning objectives with career competencies using a knowledge-based approach and an occupational skills ontology. A separate 2025 study by Pinos Ullauri and colleagues examines recommendations based on soft-skill proficiency. Together, these works show that course recommendations can use technical skills, soft skills, and career-related competencies.

8. Challenges and Safeguards

AI outputs depend on the quality of the information used to train or configure the system. Missing courses can be overlooked, and outdated descriptions can lead to misleading recommendations. A system may also repeat existing bias if its data reflects unequal opportunities. To reduce these risks, course data should be checked regularly, recommendation criteria should be documented, and students should be able to request human advice. Personal information should not be collected unnecessarily.

9. Proposed Evaluation

A pilot study could ask students to compare AI-generated suggestions with ordinary catalogue searches. Evaluation criteria could include relevance ratings, diversity of recommended courses, clarity of explanations, and the number of unsuitable suggestions. Teachers or counsellors could review whether the system correctly interprets course outcomes. The pilot should include students with different interests and backgrounds. This paper proposes these tests but reports no original experiment.

10. Conclusion

AI can support course discovery by connecting student interests with skills and learning outcomes described in course information. NLP may help interpret text, while a matching engine can rank relevant options. The system should explain its reasoning, allow users to correct their profiles, protect privacy, and leave the final choice with the student.

References

• Vo, N. N. Y., Vu, Q. T., Vu, N. H., Vu, T. A., Mach, B. D., & Xu, G. (2022). Domain-specific NLP system to support learning path and curriculum design at tech universities. Computers and Education: Artificial Intelligence, 3, 100042. https://doi.org/10.1016/j.caeai.2021.100042

• Beutling, N., & Spahic-Bogdanovic, M. (2024). Personalised Course Recommender: Linking Learning Objectives and Career Goals through Competencies. Proceedings of the AAAI Spring Symposium Series, 3(1), 72–81. https://doi.org/10.1609/aaaiss.v3i1.31185

• Pinos Ullauri, L. A., et al. (2025). A Recommender System of Postgraduate Courses Based on Soft Skills: A Psychometric-inspired Approach. International Journal of Artificial Intelligence in Education. https://doi.org/10.1007/s40593-025-00463-z